
Project Control, Review and Audit

5.1 Introduction

Project control involves a regular comparison of performance against targets, a search for the causes of deviation and a commitment to check adverse variances.

Project control serves two major functions:

1. It ensures regular monitoring of performance.
2. It motivates project personnel to strive for the achieving project objectives.

Project control is generally done by establishing the controls and controlling the ongoing activities using those controls.

Project Control involves four steps:

They are,

1. Setting targets for what should be achieved.
2. Measurement of what is happening and anticipation of what may happen.
3. Comparison between what should happen and what is happening or likely to happen.
4. Taking corrective actions to make things happen, as they should.

These four steps should follow each other till the work is completed.

5.2 Project Control Purposes

The purposes of the project control are:

1. To observe the work in progress.
2. To ensure that it follows the plan.
3. To understand the basic problems.
4. To take appropriate action when necessary.

5.3 Problems of Project Control

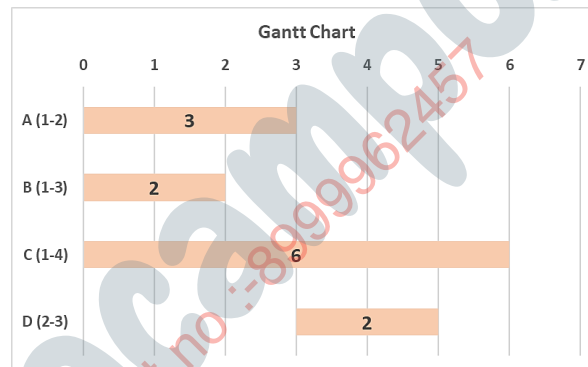
Effective control is crucial for the realisation of project objectives. There are three main reasons for poor control of projects. Viz.,

1. Characteristics of the ultra-mega projects like, complexities, maintenance of non-routine activities, co-ordination and communication problems.
2. People problems like, managers do not have required experience and training, lack of competence but have inclination to control projects.
3. Poor control and information system: Delay in reporting performance, inappropriate level of detail, unreliable information.

5.4 Gantt Charts

In dealing with complex projects a pictorial representation showing the various jobs to be done, and the time and money involved is generally helpful. One such pictorial charts is the bar chart, was developed by **Henry Gantt**. It consists of two coordinate axes, one representing the time elapsed and the other, jobs or activities performed. The jobs are represented in the form of bars.

The length of a bar indicates the duration of the job or activity.



In Fig. shown the activities A, B and C can start at the same time and proceed in parallel, though they take different time intervals for their completion. Activity D, however, cannot begin until the activity A is over. The bars representing A and D therefore run serially.

Let us consider a specific example. A piece of equipment is made of parts A and B which are to be assembled together before they are dispatched. Part A is of cast steel, which requires a pattern and a mould. Part B is a machined item made on special machine M which needs to be purchased and installed. Part A requires specialised-

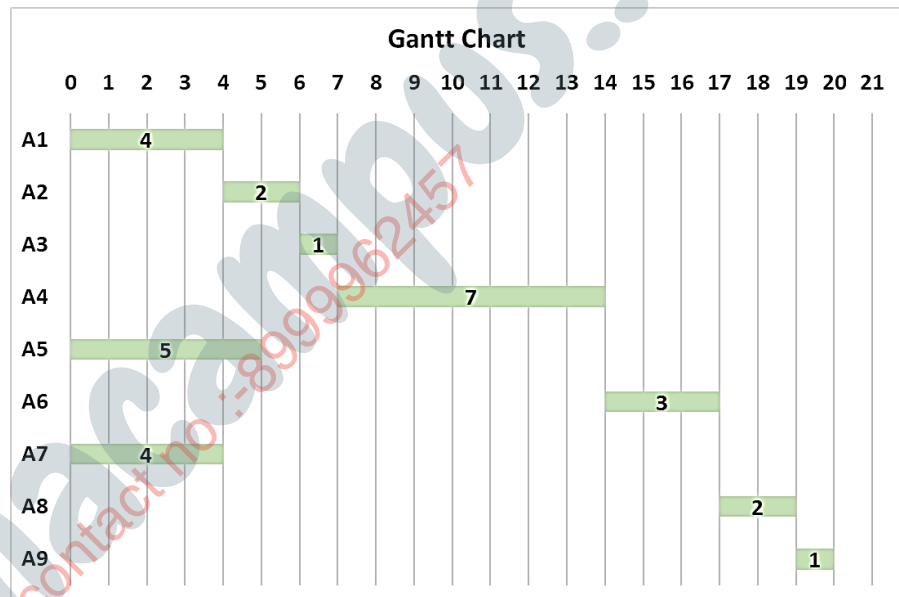
treatment before assembly. The assembly needs to be tested with a specially constructed rig before dispatch.

The time scale for each activity is as follows:

A1	Preparing a pattern for casting	4 Weeks
A2	Preparing a mould	2 Weeks
A3	Casting and cleaning operation of A	1 Weeks
A4	Heat-treatment of A	7 Weeks
A5	Machining part B	5 Weeks
A6	Assembling part A and B	3 Weeks
A7	Preparing the test rig	4 Weeks
A8	Testing the assembly	2 Weeks
A9	Packing for dispatch	1 Weeks

The Gantt chart for this project is shown below. The various activities are shown along the vertical axis and the time elapsed along the horizontal axis.

Activities	Start Day	Duration (Weeks)
A1	0	4
A2	4	2
A3	6	1
A4	7	7
A5	0	5
A6	14	3
A7	0	4
A8	17	2
A9	19	1



Weaknesses in Gantt Charts:

Gantt chart may appear to be an excellent pictorial representation of a project. However, in practice, Gantt charts have some limitations.

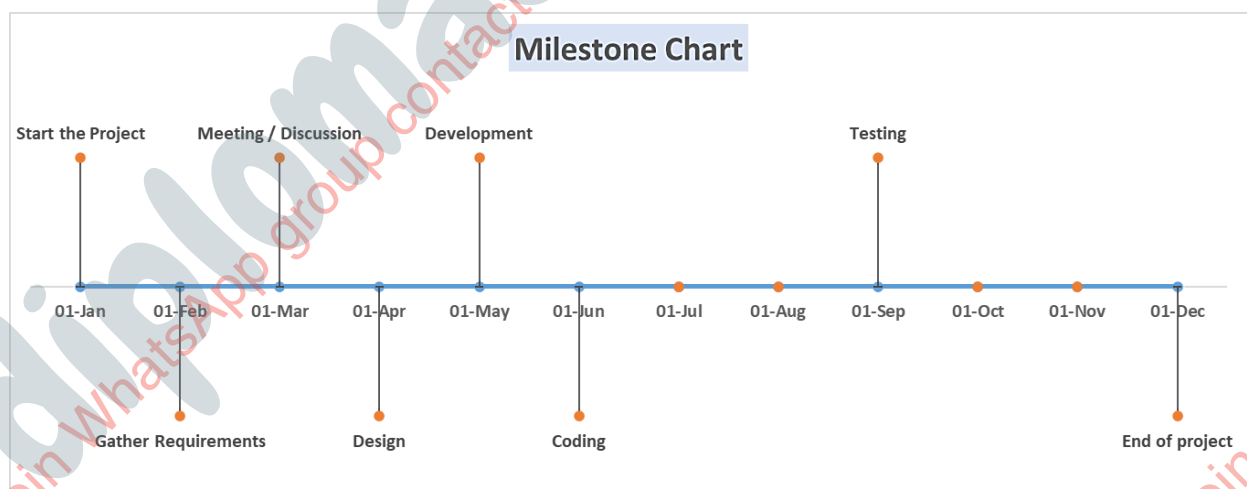
A few of these are:

1. Interdependent on activities.
2. They can become extraordinarily complex.
3. The size of the bar does not indicate the amount of work.
4. They need to be constantly updated.
5. Difficult to see on one sheet of paper.

5.5 Milestone Charts

Because of the shortcoming of the Gantt chart in meeting the requirements of the modern-day management, efforts have been made to modify it by adding new elements. This modification is called the milestone system. Milestones are key events in time, which can be identified when completed as the project progresses. In the Gantt chart a bar, which represents a long-term job is broken down to several pieces, each of which stands for an identifiable major event. Each event is numbered and an explanatory table is given, identifying the number with the event. These are specific events (points in time) which management has identified as important reference points during the completion of the project. This work breakdown increases the awareness of the interdependency between the tasks.

From Gantt chart and milestone charts two important points to be noticed are that: (a) the long-time jobs are identified in terms of specific events or milestones; and (b) these milestones or key events are plotted against the time scale indicating their achievements by specified dates.



5.6 Critical Path Method (CPM)

The critical path analysis is an important tool in production planning and scheduling. Gantt chart is also one of the tools of scheduling but it has some disadvantages, hence it is unsuitable for certain projects. The problem with Gantt chart is that the sequence of operation of a project or the earliest possible date for the completion of the project cannot be ascertained. This problem is overcome by Critical Path Method (CPM).

CPM is used for scheduling special projects where the relationship between the different parts of project is more complicated than of a simple chain of task to be completed one after the other.

According to JOHN L. BURBIDGE, one of the purposes of critical path analysis is to find the sequence of activities with the largest sum of duration, and thus find the minimum time required to complete the project. This critical series of activities is known as the 'CRITICAL PATH'.

Under CPM, the project is divided into different operations or activities and their relationships are determined and shown on the network diagram. CPM marks critical activities in a project and concentrates on them. It is based on the assumption that the expected time is actually the time taken to complete the project.

Main Objectives of CPM:

1. To find problems and obstacles in the process of production.
2. To assign time for each activity.
3. To determine the starting and finishing times of the work.
4. To find the critical path and the minimum duration of time for the project.

Advantages of CPM:

5. It provides an analytical approach to achieve project objectives.
6. It identifies most critical activities and pays more attention to them.
7. It assists in avoiding waste of time, energy and money on unimportant activities.
8. It provides a standard method for communicating project plans and schedules.

PERT (Programme Evaluation and Review Technique):

PERT is a time-event network analysis technique designed to watch how the parts of a programme fit together during passage of time and events. The special project office of the U.S Navy developed this technique in 1958.

The PERT system is based on three-time estimates of the performance time of an activity.

1. **Optimistic time (t_o):** The shortest possible time required for the completion of an activity.
2. **Most likely time (t_m):** The time taken by the activity under normal conditions.
3. **Pessimistic time (t_p):** The maximum possible time taken by the activity under worst conditions.

Expected time (t_e) or Standard average time:

The expected time of an activity is equal to the average time. This is the time that activity is expected to consume when executed.

$$t_e = \frac{t_o + 4t_m + t_p}{6}$$

Adding weight to the most likely estimate (t_m) will result in most probable outcome. The most likely estimate is weighted by a factor of 4 i.e. ($4 t_m$). It is counted four times and optimistic and pessimistic estimates one time each, yielding a total of six values. Hence, we divide by six when determining the **PERT weighted average time**.

Measures of Variability:

Variability in PERT analysis is measured by variance or standard deviation. It is used to find the probability of completing the whole project by a given date.

Steps involved in calculating standard deviation of critical path:

- (a) Determine Standard Deviation (SD) of each activity on the critical path.
- (b) Determine SD of total duration of critical path on basis of step (a).

$$\text{Standard deviation } (\sigma) = \frac{t_p - t_o}{6}$$

Where, σ = Standard Deviation, t_p = pessimistic time, t_o optimistic time.

$$\text{Variance} = (\text{Standard deviation})^2 = \sigma^2 = \left[\frac{t_p - t_o}{6} \right]^2$$

Probability of Completion by a Specific Date (Z):

With information of mean (t_e) and standard deviation (σ) for critical path duration, we can compute the probability of completion by a specified date (D) as follows:

$$Z = \frac{D - t_e}{\sigma}$$

PERT Illustration:

Find the probability of completion of the project by a specific date 20, 25 and 30 with mean time of 28 days and standard deviation of 3.07.

Solution:

Specified Date (D)	Z	Probability of Completion by D
20	$\frac{20 - 28}{3.07} = -2.6$	0.005
25	$\frac{25 - 28}{3.07} = -1.0$	0.159
30	$\frac{30 - 28}{3.07} = 0.6$	0.726

From the table it is seen that probability of completing the project in 20 days is 0.005 x 100 = 0.5 %, in 25 days 15.9% and in 30 days 72.6%.

Table 5.6: Cumulative Probability up to Z for Standard Normal Distribution

z	Cumulative probability
-3	0.001
-2.8	0.003
-2.6	0.005
-2.4	0.008
-2.2	0.014
-2	0.023
-1.8	0.036
-1.6	0.055
-1.4	0.081
-1.2	0.115
-1	0.159
-0.8	0.212
-0.6	0.274
-0.4	0.345
-0.2	0.421
0	0.500
0.2	0.579
0.4	0.655
0.6	0.726
0.8	0.788
1	0.841
1.2	0.885
1.4	0.919
1.6	0.945
1.8	0.964
2	0.977
2.2	0.986
2.4	0.992
2.6	0.995
2.8	0.997
3	0.999

Advantages of PERT:

PERT is a very important tool for planning and control. It has the following merits.

1. It provides a graphical display of project activities that helps the users to understand the relationships among the activities.
2. It is the ideal technique for planning and operational level control of projects.
3. It allows project managers to do 'what if' analysis on project activities.
4. PERT helps managers and subordinates to make a plan for production.
5. PERT concentrates attention on critical element that may need correction.
6. This system, creates a pressure for action at the right time, level and at the right event.

Limitations in Using PERT:

1. It is a time-consuming and expensive technique.
2. It is based on the assumptions hence may not always be true.
3. PERT is not suitable when a reasonable estimate of time schedule is not possible.
4. It is not useful for routine planning of recurring events.

Difference between PERT and CPM:

PERT	CPM
Stands for "Project Evaluation and Review Technique".	Stands for "Critical Path Method".
PERT is appropriate where time estimates are uncertain for activities.	CPM is good when time estimates are found with certainty.
It is concerned with events, which are the beginning or ending points of operation.	It is concerned with activities.
Suitable for non-repetitive projects.	Suitable for repetitive projects.
Can be analysed statistically.	Cannot be analysed.
PERT is not concerned with relationship between time and cost.	CPM establishes a relationship between time and cost.
It is probabilistic in nature.	It is deterministic in nature.
It can be applied only for big projects.	It can be applied for both big and small projects.
It is based on THREE-time estimates.	It is based on SINGLE-time estimate.

5.7 Construction of a Network

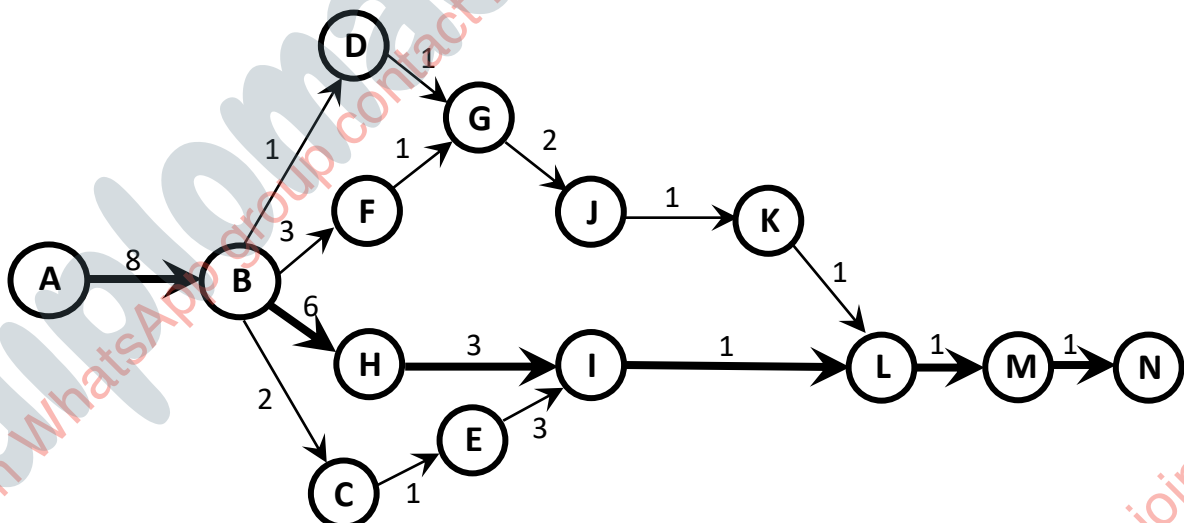
Consider a project of manufacturing a new model car. The activities involved are shown in the below table with their sequential order, duration and labour needs.

Table: New Model car - Network Activities

Activity Letter	Activity Description	Preceding Activities	Duration (days)	No. of Employees needed
A	Start	Nil	0	-
B	Design	A	8	3
C	Build frame	B	2	5
D	Build doors	B	1	2
E	Fix axles, wheels and fuel tank	C	1	3
F	Build body shell	B	3	7
G	Fit doors to body shell	D, F	1	2
H	Build and test engine	B	6	5
I	Assemble and test chassis	E, H	3	3
J	Paint body	G	2	4
K	Interior	J	1	3
L	Mount body to chassis	I, K	1	3
M	Road test the car	L	1	2
N	Finishing touch	M	1	2

For the above project we can construct a network. A network contains connected 'events' and 'activities'. 'Event' refers to the 'starting' or 'completion' of specific jobs. The 'circles' in the network indicate the 'events'. The circles are known as 'nodes'. 'Activity' refers to progress of work leading from one event to another. This is indicated by the 'arrows' in the network.

Fig. Network diagram - Car Project:



Time scheduling using network diagram:

Event A is the starting point. B indicates the completion of 'design' work. A→B, indicates the progress of design activity, which needs 8 days for completion.

Until event B is completed activities C, D, F and H cannot be started. Hence, all these activities have common start node, viz. B. Activity G cannot be started, until activities D and F are completed. Hence, activities D and F converge at G.

Now, the different routes of the network can be deduced. The routes of the project are:

Route I : A-B-D-G-J-K-L-M-N

Route II : A-B-F-G-J-K-L-M-N

Route III : A-B-H-I-L-M-N

Route IV : A-B-C-E-I-L-M-N

The time duration of the different route is,

Route I involves: $8+1+1+2+1+1+1+1 = 16$ days

Route II involves: $8+3+1+2+1+1+1+1 = 18$ days

Route III involves: $8+6+3+1+1+1 = 20$ days

Route IV involves: $8+2+1+3+1+1+1 = 17$ days

The longest route is: **A-B-H-I-L-M-N** which takes 20 days. This route is called the critical path. The network diagram represents the critical path by thick arrows to indicate the critical path.

Time required for completion of the project is given by the time duration of the critical path. If everything goes well, by end of 20th day the new model car would be ready. For this to happen, each activity in the critical path has to be taken up and completed as per schedule.

The concepts of **Earliest Start Time (EST)**, **Earliest Finish Time (EFT)**, **Latest Start Time (LST)** and **Latest finish time (LFT)**, slack and free slack can be calculated. The slack refers to the extent, an activity can be delayed without affecting completion of the project. This is given by **LST – EST** or **LFT – EFT**. Free slack refers to spare time available for an activity when all succeeding activities started at their respective **EST**.

Table: EST, EFT, LST and LFT

ACTIVITY	EST	EFT	LST	LFT	Total Slack (LFT-EFT) Or (LST-EST)	Free Slack
A-B	1	8	1	8	0	0
B-D	9	9	13	13	4	0
D-G	10	10	14	14	4	2
B-F	9	11	11	13	2	0
F-G	12	12	14	14	2	0
G-J	13	14	15	16	2	0
J-K	15	15	17	17	2	0
K-L	16	16	18	18	2	2
B-H	9	14	9	14	0	0
H-I	15	17	15	17	0	0
B-C	9	10	12	13	3	0
C-E	11	11	14	14	3	0
E-I	12	14	15	17	3	3
I-L	18	18	18	18	0	0
L-M	19	19	19	19	0	0
M-N	20	20	20	20	0	0

Uses of Network Techniques:

1. The graphic representation shows how each activity is dependent on others and helps in better scheduling, monitoring and control of project activities.
2. To prepare the network diagram, considerable planning, analysis and in-depth evaluation of the whole project are needed, which in turn help in better execution of the project.
3. Network techniques can serve as indicators of bottle necks and potential trouble spots and this helps in handling of pitfalls, so that the project will progress as per original plans.
4. Network diagram illustrates the coordination required among several project teams.
5. Network diagram helps in identifying critical paths and thereby helps diversion of resources to them so that they are not lagging behind schedule.
6. Network techniques help in resource allocation.
7. Network techniques help in cost control.

Resource Allocation and Levelling Through Network

PERT and CPM techniques are not simple statistics involving the computation of times, EST, LST, EFT, LFT and drawing the diagram. It is a dynamic tool. It helps in resource allocation and resource levelling.

Resource allocation means how much resources are to be diverted to the project concerned day after day as the project progresses.

Progress of activities and labour needs for EST

DAYS	B	C	D	E	F	G	H	I	J	K	L	M	N	TOTAL LABOUR
1	3													3
2	3													3
3	3													3
4	3													3
5	3													3
6	3													3
7	3													3
8	3													3
9		5	2		7		5							19
10		5			7		5							17
11				3	7		5							15
12						2	5							7
13							5		4					9
14							5		4					9
15								3		3				6
16								3						3
17								3						3
18											3			3
19												2		2
20													2	2
TOTAL LABOUR DAYS PER ACTIVITY	24	10	2	3	21	2	30	9	8	3	3	2	2	119

In the table, the number of workers required for each activity is given. With that information we can compute daily requirement of human resource.

The requirement depends on when the activities are scheduled to begin, ie. as per their EST or LST or any intermediary time. Suppose the activities are scheduled as per EST, then the human resource needs would be as in the table.

From the table, we get requirement of labour on day-to-day basis, (see last column) and how much is required activity-wise (see last row).

Progress of activities and labour needs for LST

DAYS	B	C	D	E	F	G	H	I	J	K	L	M	N	TOTAL LABOUR
1	3													3
2	3													3
3	3													3
4	3													3
5	3													3
6	3													3
7	3													3
8	3													3
9							5							5
10							5							5
11					7		5							12
12		5			7		5							17
13		5	2		7		5							19
14				3		2	5							10
15								3	4					7
16								3	4					7
17								3		3				6
18											3			3
19												2		2
20													2	2
TOTAL LABOUR DAYS PER ACTIVITY	24	10	2	3	21	2	30	9	8	3	3	2	2	119

In the same way, we can find what would be the daily labour needs when the works are scheduled as per their LST, the above table gives the same.

Between 9th and 17th days, there are differences in the daily needs under the two patterns of job scheduling. During the period 1st to 8th day only one critical activity is in operation and during 18th to 20th days too only critical activities are operated. So, under both scheduling patterns the labour needs are same.

Resource levelling means distributing the daily resource needs to the extent possible. We find that under the EST scheduling the labour needs on 9th through 17th

days are respectively 19, 17, 15, 7, 9, 9, 6 and 3 persons and under the LST scheduling, labour needs are 5, 5, 12, 17, 19, 10, 7, 7 and 6 persons per day respectively.

The daily needs are highly varying. Certain levelling or smoothening or reducing the variations in daily needs can be attempted. This is called as resource levelling.

Why Resource levelling is needed?

The reasons are:

1. Resource constraint can be one of the reasons. Say, not more than 12 persons are available on any one day, whereas we need as much as 19 persons on one day. By rescheduling noncritical activities using their slack times, the above purposes can be served.
2. Practically speaking, too much needs on some days and too little needs on other days are not signs of good planning.
3. The disruption in work is more probable when there is peaks and troughs in resources.
4. Optimum utilisation of permanent/owned facilities and avoiding idle time are possible with resource levelling exercises.

Say in our case only 12 labours are available on any given day. Is it possible to complete the project on time with only 12 persons? May be some rescheduling can be thought of.

Let us go by LST scheduling given in table for activities D and F. Let us advance F by 3 days slack. That is from 9th to 11th.

By doing so, the labour needs on 9th, 10th, and 11th day go up to 12 persons. And for 12th and 13th days go down to 10 and 12. Hence maximum labour availability condition will satisfy.

Further, by advancing D by a day to 12th and the labour needs will become 12 and 10 for 13th day. Now the labour requirements from 9th to 14th will become 12, 12, 12, 12, 10 and 10 persons. This arrangement also ensures better utilisation of permanent employees say 9 persons.

Progress of activities and labour needs for LST with resource levelling

DAYS	B	C	D	E	F	G	H	I	J	K	L	M	N	TOTAL LABOUR
1	3													3
2	3													3
3	3													3
4	3													3
5	3													3
6	3													3
7	3													3
8	3													3
9					7		5							12
10					7		5							12
11					7		5							12
12		5	2				5							12
13		5					5							10
14				3		2	5							10
15								3	4					7
16								3	4					7
17								3		3				6
18											3			3
19												2		2
20													2	2
TOTAL LABOUR DAYS PER ACTIVITY	24	10	2	3	21	2	30	9	8	3	3	2	2	119

5.8 Network Technique in Project Scheduling

Network technique is commonly used in project scheduling. When will each activity be commenced? when the same has to be completed? which activity can be delayed? when will the project be completed? Such related questions are answered by PERT and CPM techniques. All these require time estimates and sequential relations between jobs.

Time estimates are made based on past experience, the job nature and availability of resources. In PERT, 3 estimates of time for each activity and the expected time are calculated. In CPM only one time estimate is made, but estimates may be revised in both the methods.

The sequence of operations is to be known thoroughly. Once time estimates and sequential relations are known, activities scheduling can be prepared.

Under PERT we can find the probability of finishing the project by certain date. For this we need to know the standard deviation of activity times for critical activities.

Calculate the standard deviation for each of the critical activities. Square each of the standard deviation values. Add the squared values. Take square root for the summated value. This is taken as the project standard deviation of the earliest finish time.

$$\text{Standard normal variate (Z)} = \frac{\text{Due date} - \text{Expected date of completion}}{\text{Project standard deviation}}$$

We may find the probability of completion by 22nd day from commencement for our car project, given the expected completion by 20th day.

We need to know the three time estimate for the critical activities which are as follows:

Activity	t_p	t_m	t_o	$\sigma = (t_p - t_o)/6$	Variance
B	10	9	2	$(10-2)/6 = 1.33$	1.77
H	9	4.5	3	$(9-3)/6 = 1$	1
J	5	3	1	$(5-1)/6 = 0.66$	0.44
L	5	3	1	$(5-1)/6 = 0.66$	0.44
M	5	1.5	1	$(5-1)/6 = 0.66$	0.44
N	5	1.5	1	$(5-1)/6 = 0.66$	0.44

$$\text{Standard deviation of project} = \sqrt{1.77 + 1 + 0.44 + 0.44 + 0.44 + 0.44} = 2.128$$

$$Z = (22-20)/2.128 = 0.9375$$

Area under normal curve corresponding to $Z = 0.9375$ is equal to $= 0.825$. That is, there is a probability of 0.825 or 82.5% that the project would be completed by 22nd day.

The probability computation is helpful in project rescheduling. If the probability is very small, there is a need for, speeding up the work by allocating more resources or postponing the due date.

5.9 Crashing Project Duration through Network

Sometimes a project has to be completed sooner than the planned time. In the above example the project duration is 20 days. Suppose, if we want to complete the project in 15 days, it may be possible by allocating extra resources. Extra cost may arise due to work overtime.

The below table gives the normal time & cost and crash time & cost for the activities of car project. The normal cost is ₹ 1,19,000 and the crash cost is ₹ 1,55,750. But we can complete this in 15 days, by proceeding methodically with a cost less than ₹ 1,55,750.

Table: Normal and Crash time / cost

ACTIVITY	NORMAL		CRASH		COST REDUCTION PER DAY
	TIME	COST	TIME	COST	
B	8	24000	6	30000	3000
C	2	10000	1	11000	1000
D	1	2000	0.5	6000	8000
E	1	3000	0.5	6000	6000
F	3	21000	2	22000	1000
G	1	2000	1	2000	-
H	6	30000	4	36000	3000
I	3	9000	2	15000	6000
J	2	8000	1	9250	1250
K	1	2000	0.5	6000	6000
L	1	3000	0.5	6000	6000
M	1	2000	0.5	4500	5000
N	1	2000	1	2000	-
		119000		155750	

Find the activities in the critical path where time can be cut with minimum extra cost. The goal is to reduce project time with the least increase in project cost. Work out the cost of crash per day for each activity.

$$\text{Cost of crash per day} = \frac{(\text{Crash cost} - \text{Normal cost})}{(\text{Normal time} - \text{Crash time})}$$

Route-III is the critical path. Activity N cannot be crashed. And only activities B, H, I, L and M can be crashed. B or H is the least crash-cost-per-time activity. Consider activity B for crashing by TWO days.

The extra cost for two days is $2 \times 3,000 = ₹ 6,000$. Now route III and route II are critical as both have 18 days duration. To cut project duration we have to reduce the duration of both the routes.

Route I : A-B-D-G-J-K-L-M-N Route I involves: $8+1+1+2+1+1+1+1 = 16$ days

Route II : A-B-F-G-J-K-L-M-N Route II involves: $8+3+1+2+1+1+1+1 = 18$ days

Route III : A-B-H-I-L-M-N Route III involves: $8+6+3+1+1+1 = 20$ days

Route IV : A-B-C-E-I-L-M-N Route IV involves: $8+2+1+3+1+1+1 = 17$ days

In route II the crushable activities are F, J, K, L and M. Activities L and M are common in route III. So, consider the least crash cost activity that is F.

In route III the least crash cost activity is H. By crashing F & H by one day each, the project duration becomes 17 days and extra cost is $1,000 + 3,000 = ₹ 4,000$.

Now route II, III & IV become critical paths with a duration of 17 days. To reduce project duration, crashing has to be done in all these three routes.

The crushable activities in route-IV are: C, E and I. But 'I' is common activity with routes II and III. Select C for crashing as it is least cost per day activity. In route-III, activity H is crushable by a day with extra cost 3,000. In route-II, 'J' can be crashed by a day with extra cost of ₹ 1,250.

By doing so, project duration is cut by a day to 16 days at an extra cost of ₹ 5,250. Now all the four routes are in critical with 16 days duration each. We want to reduce the project duration to 15 days ie. the duration of each of the 4 routes has to be cut by a day.

This can be done by crashing L and M at an extra cost of $3,000 + 2,500 = ₹ 5,500$. So, the extra cost of crashing is **6,000 for 2 days + 4,000 for 1 day + 5,250 for 1 day + 5,500 for 1 day = ₹ 20,750**. Hence, the crash cost of the project is only 20,750, more than normal cost. Hence, the project can be crashed and the duration is reduced to 15 days from 20 days.

Parallel and Serial Scheduling:

The estimates of activity duration are approximate. Reality varies from the estimate. For the optimum use of time and computer resources levelling methods are used like a "parallel" or a "serial" scheduling.

Parallel Scheduling:

This can be visualised as follows. Imagine a network with several possible start points. This network is to be schedule on a day-by-day basis through the project. First, we consider day one.

Take all the start activities of the network. Those are all the activities, which could start at day one. Build an 'eligibility list' consisting of those activities. Take the activity with the highest priority from the eligibility list. Can it start yet? Is its earliest start less than or equal to the day we are considering? If so, compare its resource requirements for its next day with the resource availability on day one of the project. If all relevant resources are available in sufficient quantity, schedule the first day of that activity to occur on the first day of the project.

If that completes the activity, then include all successor activities into the eligibility list. Repeat the process for the activity with the next highest priority from the eligibility list, and so on until eligible activities have been considered for that day.

Then move to the next day of the project. Any amounts remaining of 'pool' resources are rolled over to day two. The actions described in the above paragraph are repeated for day two scheduled.

Notice that this process works more naturally when the resources available to a project are limited, and the time available to complete it can be extended indefinitely.

Serial Scheduling:

Serial scheduling considers each activity in turn, rather than each day of the project in turn. It can be visualised as follows. Consider a two-dimensional table of resource availabilities. It will have an amount available for every resource for every day of the project. Against this there is a pre-sequenced list of activities, which comprise the projects. In addition to its time analysis results, each activity will have an 'earliest feasible start' figure, which will initially be the same as its earliest start.

Each activity is taken from the list of activities in turn. The section of the resource tables between the activity earliest feasible start and the activity late finish is scanned to see if the activity can be scheduled as a whole. If so, it is scheduled at the first available point.

If not, then if the activity is splittable, an attempt is made to fit the activity in between its earliest feasible start and latest finish in sections. If this is impossible, an extra layer of resource availability is called for any resource which was in inadequate supply, and the process is repeated for the activity, until it can be fitted in.

If an activity is scheduled at a point later than its early start, the earliest feasible start of all successor activities is updated to the greater than the finish of the activity that has just been scheduled. When all this has been done for every activity in the network, the project has been scheduled.

Notice that this process works more naturally if there is a fixed end date to the project, but the resources can be exceeded if necessary.

Serial methods are now more popular than parallel methods, for a number of reasons. Parallel scheduling tends to split activities rather more than serially, a parallel scheduling algorithm would like to be able to split any activity into one-day sections, particularly if activities have complex resource requirements, or if resource availability changes during the project.

Parallel schemes are typically heavy users of computer resources, both in the time taken to carry out a schedule, and in the amount of computer memory required per activity (which can limit the size of network that can be calculated on a given computer).

Special Features in Resource Levelling

The above descriptions of the parallel and serial techniques show the basic method involved in each case. There are many refinements and special features, which can be added to both processes.

A 'threshold' amount can be associated with each resource, as mentioned above. This is an additional emergency allocation of the resource which can be used if the projects are about to run behind time. The threshold resource level cuts in when activities would otherwise be delayed past their latest finish.

Serial levelling schemes sometimes have a feature whereby two project ends dated are specified-a desired project end and a maximum project end. Resources are classified into two sections 'important' and 'exceed'. If the scheduling system would have exceeded the availability of an 'important' resource, (and its threshold level, if there was one), then the activity is allowed to delay itself past its latest finish. It is not allowed to delay past its secondary latest finish- that is, the latest finish relative to the maximum project end date.

Key words used: Project construction, CPM, PERT, Traffic schedule, Common network, Parallel and serial scheduling, Crash time, Time estimate.

5.10 Project Review

Project review is the final phase of Project Management. It is an important aspect in the process of planning to correct the errors and rectify the same.

Review is an important tool to identify the shortcomings during the entire implementation period. It will help us to develop or initiate corrective actions to improve the delivery and administration.

Review/Evaluation process essentially involves some important steps viz. Data collection, estimation of cost and profitability of the project. Data collection may be made both from primary sources as well as from secondary sources. The primary sources like the industry and market and the secondary sources like the published documents viz. company balance sheets, government publications, publications of independent research bodies or industrial associations.

Once the data is collected, it is necessary to shift the same for eliminating the irrelevant and retain only the significant information. On the basis of the data collected an estimate may be prepared. A comparison of the estimated and required profitability will form the basis of selection or rejection of the project.

Various phases of Project Review:

1. Initial review
2. Performance evaluation
3. Abandonment analysis
4. Behavioural issues in project abandonment
5. Administrative aspects of capital budgeting
6. Evaluating the capital budgeting system of an organization

5.11 Initial Review

It is the first stage in the project review process. The initial review of a project is of two types:

1. Control of project in progress
2. Post audit

Control of Project in Progress:

The expenditure authorisation for a project generally specifies how much can be spent by whom and when. A periodical control is exercised during project

implementation to ensure that the actual expenditure does not deviate from the budget.

Post Audit

An audit of a project after it has been commissioned is called as post audit. Such an audit compares actual performance and planned performance. Post audit assures proper evaluation of the project.

5.12 Performance Evaluation

The performance evaluation is done periodically to measure the performance of the project. Performance evaluation may be done in terms of Economic Rate of Return (ERR) or Book Rate of Return (BRR) on investment.

Economic Rate of Return (ERR): It provides a single metric showing how a project's economic benefits compare to its costs. An ERR provides a convenient metric, produced from a cost-benefit analysis, that compares the economic costs and benefits of a program.

$$\text{Economic rate of return for a given year} = \frac{\text{Cash flow} + \text{Changes in present value}}{\text{Present value at the beginning of the year}}$$

Book Rate of Return (BRR): This is a rate of return measured based on accounting earnings. It is defined as the ratio of book income to book value.

Book income is the amount of income a company publicly report on its financial statements to shareholders. **Book value** is equal to the cost of carrying an asset on a company's balance sheet.

$$\text{Book rate of return for a given year} = \frac{\text{Cash flow} + \text{Changes in book value}}{\text{Book value at the beginning of the year}}$$

5.13 Abandonment Analysis

A project is analysed on the assumption that the firm will operate it for a given period. However, there may be chances to abandon the project before this period. The possibility of abandonment, when considered explicitly in project analysis, may change the decision itself.

A basic rule of capital budgeting says that investment decisions should be guided by the net present value criterion. The 'continuation versus abandonment' decision

will be based on the rule that the project must be abandoned if the net present value associated with abandonment is greater than the net present value associated with continuation.

5.14 Project Audit

Professional management of project needs a methodology to carry out a regular check to find whether the project is progressing as scheduled, in scope and in time.

Project audit is a formal and systematic examination of the performance of an ongoing project. It involves measurement against predefined and relevant standards. It supports management in decision-making. The corrective action taken on the observations are final process of the project audit.

5.15 Objectives of project audit

The objectives of project auditing are;

1. Creating awareness among the project staff about the problems that are likely to be encountered in completing the project and producing quality products, in planned volume and at competitive costs.
2. Providing a clear picture of the actual status of the project from time to time.
3. Prompt identification of the factors that might cause product quality problems.
4. Prompt identification of the factors that might cause or lead to time and/or cost overruns.
5. Timely spotting the problems that are associated with execution of projects and suggestions to overcome the problems.
6. Enabling the creation of a good information base for a proper estimation and costing of the project.
7. Assisting in the establishing of appropriate standards and systems and recommending suitable work techniques.
8. Identifying the specific training needs with reference to the project tasks.
9. Formalising the experience and expertise in project management to provide consultancy services to other enterprises.

5.16 Functions of Project Auditor

1. Plan and manage assigned audit projects according to established standards.
2. Oversee auditing for operational, financial and compliance areas.

3. Analyse root causes of control inefficiencies and recommend corrective actions.
4. Measure, confirm, investigate, and report the status of a project with a view of reducing the uncertainties.
5. Give advice to make recommendations.
6. Evaluate the contract base lines and give judgement on their adequacy.
7. To derive conclusions based on the audits conducted, maintain the documents related to the audit work.

5.17 Project Audit Programme

The project audit programme involves the identification and analysis of the set of contractual documents. These documents collectively define the extent and details of the project's obligations, determination of additional resource allocation, pricing and costing assumptions and the development of the detailed audit programme.

The project audit programme will have the following steps:

1. Preliminary examination of the project's organisation, administration, record keeping, controlling, planning and working methods.
2. Preparing the statements of project's current and future status.
3. Giving a detailed list of completed work as compared with the project's performance baseline.
4. Recording the costs expended by the project team to carry out the work.
5. Comparing the fulfilment of contractual obligations against commitments.
6. Evaluation of the quality of work done by the project team.
7. Recording observed facts on project planning, project organisation and staffing.

5.9 Difficulties in Establishing Audit Purpose and Scope

The purpose of project audit is to clarify the state of the project for the project management. It is a part of quality control effort. The auditor could often be misled by wrong information or out of date or incorrect documentation. He should have a system of cross verification and cross checks. The auditor will have to report the resource mismatch indicating their implications for the project. Difficulties might arise in identifying the baseline contract or work statement. Work could be carried out on the basis of verbal agreements without proper contract documentation. Absence of deliverable specifications in the contract baseline and inadequacies in delivery procedure or acceptance criteria are often problems requiring special attention. The auditor will have to resolve such ambiguities.

Questions:**Remember:**

1. Define project control.
2. What is the purpose of project control?
3. List the problems of project control.
4. What is Gantt chart?
5. List the objectives of CPM.
6. List the advantages of CPM.
7. What is PERT?
8. List the advantages of PERT.
9. List the uses of network techniques.
10. What is project review?
11. List different stages of project review.
12. What is initial review?
13. What is post audit?
14. What is abandonment analysis?
15. List the objectives of project audit.
16. What are the functions of project auditor?
17. What are the steps involved in project audit programme?

Understanding:

1. Explain the steps in project control.
2. Explain the Gantt chart with example.
3. Explain network diagram with illustration.
4. Explain the Milestone chart.
5. Explain CPM.
6. Explain PERT.
7. Differentiate between PERT and CPM.
8. Explain the difficulties in establishing project purpose and scope.
9. Explain the functions of Project Auditor.
10. Explain the difficulties in the Project Audit.
11. Explain the steps involved in project audit programme.